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Role of Hypothalamic A11 Dopamine-Histamine Circuit in Controlling Wakefulness

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Abstract:

Previous research from our laboratory shows the hypothalamic A11 region contains twice the proportion of wake-on dopamine cells than other previously studied dopaminergic nuclei. Some well-known drugs work via dopamine and histamine systems, but the circuitry remains unknown Additionally, studies suggest the role of histamine cells in the tuberomammillary nucleus (TMN) to be involved in wakefulness and speculate these cells to be able to receive dopaminergic inputs due to their expression of dopamine receptors. The purpose of this research is to investigate the interaction between the A11 and the TMN regions and assess their circuit in promoting wakefulness in mice. Here we show using in-vivo optogenetics and immunohistochemistry that dopaminergic A11 dopaminergic dendrites in the TMN contribute to maintaining wakefulness. Activation of the A11 to TMN circuit increases wakefulness by prolonging individual wake episodes. We also identify that the A11 projections activate histamine cells of the TMN. Furthermore, we show that this activation may have a strong enough effect to overcome physiological pressures to sleep. This circuitry provides a new way of understanding the dopamine and histamine co-contributions to maintaining wakefulness.

Introduction:

Wakefulness and sleep are both important regulatory states in animal physiology. Almost every active behavior, such as foraging, grooming and eating have one thing in common, they all occur exclusively during wakefulness (MG, 2011). Wakefulness is a behavioral state where animals maintain the ability to perceive environmental stimuli & develop an adaptive response to ensure survival through various levels of arousal (van Schie et al., 2021). Arousal is a functional unit of wakefulness defined as the ability to be sensitive to “potential” change in one’s environment (Rial et al.,2023). Some activities require high levels of arousal meaning high muscle and brain activity, like fleeing from danger, while others require relatively lower levels like lying down (Brown et al., 2012). Wakefulness can be electrophysiologically defined in mammals by high motor tone and activity as well as low amplitude high frequency cortical activity containing high theta waves accompanied by gamma, alpha and beta waves (Furutani, 2025). The underlying question for a long time has been “How are these physiological states (sleep & wakefulness) generated?”

Many neurotransmitters have been shown to contribute to promoting wakefulness, such as dopamine and histamine. However, it is suggested that regulation of wake & sleep and the transition between the states must be driven by multi-neurotransmitter circuits (Holst & Landolt, 2022).

Multiple studies have indicated the role of dopamine in wakefulness. For example, optogenetic stimulation of VTA-dopaminergic neurons causes a rapid awakening in anesthetized mice (Taylor et al., 2016). On the other hand, optogenetic inhibition of the VTA dopaminergic neurons induces sleep from wakefulness, suggesting that dopamine is necessary for keeping animals awake (Eban-Rothschild et al., 2016). Other experiments show that dopamine-deficient mice lacking the enzyme tyrosine hydroxylase have a significant reduction in their time spent awake (Kashiwagi et al., 2021). Conversely, mice lacking the dopamine reuptake channel (DAT) show prolonged wakefulness due to increased synaptic dopamine, suggesting that dopamine levels in the synapse must be regulated for maintaining appropriate levels of arousal (Wisor et al., 2001). Furthermore, wake-promoting anti narcoleptic drugs such as amphetamines and modafinil increase synaptic dopamine levels by blocking DAT, and again, knocking out DAT cause mice to become unresponsive to the effects of stimulant drugs (Wisor et al., 2001). Together, these findings highlight the contribution of dopamine and its importance in wakefulness control.

Research on well-known dopaminergic nuclei, such as the ventral tegmental area (VTA) and the substantia nigra (SN) show their projections onto arousal promoting nuclei such as the locus coeruleus, dorsal raphe (Monti and Monti 2007) and the striatum (Oades and Halliday, 1987). However, despite these findings, the main dopaminergic circuit responsible for maintaining wakefulness has been speculative due to other findings, which show the dopaminergic neurons of the VTA and SN not changing their firing rate across the sleep-wake cycle (Miller et al. 1983). This leaves room for a novel dopamine circuit which could potentially be responsible for promoting wakefulness.

We still do not fully know which dopaminergic circuit controls wakefulness. However, a potential brain region which could be responsible for the promotion of wakefulness is the dopaminergic A11 nucleus of the caudal hypothalamus which is the main source of dopamine in the spinal cord and is involved in modulation of the "*clock*" gene expression and the rhythm of the circadian clock which is expressed maximally as wakefulness begins (Piña-Leyva et al., 2022). Additionally, optogenetic activation of the A11 region in mice shifts their behavior towards grooming and stretching which require higher levels of arousal as measured by motor activity (Koblinger et al., 2018).

Previous work from the Peever lab shows that the A11 dopamine neurons are "wake-on "cells. Sleep deprived mice were sacrificed and stained for the cell activity marker c-Fos which shows that the A11 contains almost double the proportion of these wake-on neurons compared to other wake promoting regions. Additional data from the Peever lab shows that optogenetic activation of the A11 promotes arousal. Taken together, these findings suggest a wake promoting role in the hypothalamic A11 region, however the downstream effectors from this region remain unknown.

Another wake-promoting neurotransmitter is histamine, which is produced exclusively by neurons of the tuberomammillary nucleus (TMN) (Monti, 2011). These neurons are wake-on (Takahashi et al., 2006) and innervate several brain regions including the cortex and thalamus. Antihistamine medications and histamine receptor antagonists cause sleepiness and mice lacking the enzyme which synthesizes histamine (histidine decarboxylase) cannot maintain wakefulness (Parmentier et al., 2002 & 2016). Moreover, increasing levels of synaptic histamine have been shown to increase wakefulness (Kalivas, 1982). Although we know that histamine cells can modulate wakefulness, we do not know which neurons communicate with them.

Interestingly, the wake-promoting drug modafinil acts as a blocker of dopamine reuptake channels as previously stated. However, more recent studies show that ablation of histamine cells in mice reduces the effects of modafinil and causes fragmented sleep (Yu et al., 2019). This suggests that the effects of this drug which act on the dopaminergic system are in part mediated through histamine neurons. There is also evidence from the Peever lab in support of the TMN receiving dopamine input coming from the A11 which could modulate its wake-promoting effects. Surprisingly, the more famously studied dopaminergic nuclei, the VTA and SN do not project to the TMN region (Arbuthnott & Wright, 1982). Collectively, these findings indicate that the wake-promoting effects of the TMN are regulated by dopaminergic input which may originate from the A11.

We know that the A11 sends dopaminergic input to TMN, and both regions are known to be involved in wake promotion. Hence, we hypothesize that activation of the A11 dopaminergic inputs to the TMN will promote wakefulness from sleep and this effect is generated by the activation of histamine cells by the A11 projections.

Methods:

Animals and stereotaxic viral injections

Adult mice, 6-12 weeks of age were used to optogenetically excite dopamine projections from the A11 to TMN (n=21 (all groups). We used 14 Th-Cre mice which express the Cre-recombinase enzyme under control of the gene for tyrosine hydroxylase (the necessary enzyme to make dopamine), to restrict the recombination of this enzyme in dopaminergic neurons.

To activate the A11 inputs to the TMN, we expressed ChETA- eYFP in the dopamine neurons of the A11 using a Cre-dependent adeno associated virus: AAV5-EF1α-DIO-ChETA-eYFP (200 nL). We bilaterally injected the viral vector into the A11 ((-1.94 AP, +/- 0.35 ML, 4.6 DV) mm from the Bregma). The excitatory opsin channel ChETA is transcribed in dopaminergic neurons as previously outlined via Th-cre mice. Expression of eYFP reporter gene allows the labelling of

A11 terminals in the TMN. Our control animals will be injected with AAV5- EF1α-DIO-eYFP to ensure the presence of foreign proteins and surgical injections to the brain does not influence sleep-wake states. Following the surgery, in allowing for the vector expression in the brain, mice are left to recover for 2 weeks in their home cage.

EEG/EMG and Optical fiber implants

Following the recovery period after bilateral injection of the AAV, mice underwent a second surgery to implant head plugs containing electroencephalogram (EEG) and electromyogram (EMG) electrodes for cortical and muscle recordings, and optic fibers for bilateral optogenetic stimulation of the A11 projections to the TMN. The head plugs and electrodes were custom made in-house by making 4 EEG screws and 4 EMG wires and soldering them onto a microstrip connector. We used insulated multi-stranded steel wire for making both EEGs and EMGs. For EEGs the insulation layer was stripped from one end of the wire and the soldered to a steel screw which was sanded down to allow for implanting onto the frontal and parietal bones, while the other end was soldered onto the head plug base. For EMGs, the insulation was stripped from one end and a loop was formed, allowing for the electrode to be sutured into the muscles of the masseter and neck during surgery while the opposite end is soldered onto the head plug base. During surgery, 4 holes were drilled onto the skull of the mice while under anesthesia for the placement of the EEG screws. Next, the EMG electrodes were sutured underneath the skin onto the mentioned muscles. The base of the head plug housing the electrodes and optic fiber was secured on the head of the mice using a mixture of dental cement. The 2 optic fibers were made in house and placed bilaterally above the A11 terminals in the TMN ((-2.3 AP, +/- 0.9 ML, -5.1 DV) mm from Bregma). After the surgery mice are left to recover for an additional 14 days prior to conducting behavioral tests.

Sleep-wake monitoring and optogenetic experiments

After the recovery period, mice are left to habituate to their cage for 3-5 days. To assess the wake promoting effects of the A11 to TMN projections, behavioral recording will be conducted from 2pm-5pm during light phase and from 8pm-11pm during dark phase over a 3-day period. 20 Hz blue laser pulses of 473nm were delivered during sleep and wakefulness to activate the A11 projections in the TMN through activating the excitatory opsin channel ChETA. The firing frequency of 20 Hz was used to mimic the activity of the dopamine neurons shown to illicit behavioral change as outlined by previous research (Tsai et al., 2009). Mice underwent an open-loop excitation, where laser pulses were delivered for 10s/min for180 minutes regardless of the state of animal. After behavioral experiments are complete mice will be untethered and perfused. To assess if A11 dopamine neurons activate histamine cells, the same surgical process was followed but laser pulses were delivered 10 seconds on and 10 seconds off for 1 hour. After 90 minutes following this activation, mice will be untethered and perfused.

Sleep deprivation

Mice undergoing sleep deprivation were kept in their home cage (n=5). Using a paintbrush, mice were gently stroked on their back when visually appearing to fall asleep to trigger arousal to increase physiological sleep pressure for 2 hours during the light phase. This method of sleep deprivation has been previously used in various studies and has been shown to produce physiological sleep pressure (Carter er al., 2009) without increasing stress in mice (Palchykova et al., 2006).

Brain extraction and slice preparation

Following the behavioral recording and laser stimulations, mice were intracardially perfused with paraformaldehyde and their brains were extracted and cryoprotected in a solution of 4% paraformaldehyde and 30% sucrose for 2 days before being frozen at -80°C. Brains were sliced into 40 µm layers using a cryostat machine.

Immunohistochemistry and microscopy

Immunohistochemistry (IHC) was performed on brain tissue in a solution of 0.1M phosphate-buffered saline (PBS). To verify the location of AAV5-EF1α-DIO-ChETA-eYFP injection sites and placement of the optic fiber, brain slices were stained for histamine cells as well as the reporter protein within the viral construct. We incubated brain slices in chicken anti GFP (will in this case bind to eYFP) and rabbit anti HDC primary antibodies for 24 hours at 4°C. We then re-incubated the slices in Alexa-488 goat anti-chicken and Cy3 goat anti-rabbit secondary antibodies. Secondary antibodies contain fluorescent tags which bind to their corresponding primary antibodies, leading to fluorescent marking of target neuronal markers. To determine activation of histamine cells of the TMN by input from the A11, brain slices were stained for histamine cells, the reporter protein of the viral construct and a marker of neuronal activity. Slices were incubated in guineapig anti-c-Fos (marker of neuronal activity), rabbit anti HDC and chicken anti GFP primary antibodies for 24 hours at 4°C. Slices were then re-incubated in Alexa-647 goat anti-guineapig, Cy3 goat anti-rabbit and Alexa-488 goat anti-chicken secondary antibodies. All brains were washed in 0.1M PBS 3 times for 5 minutes to wash any excess/unbound antibodies. In the second wash, antibodies for DAPI were added to stain the cell's nucleus to better visualize the cells. Cells stained for HDC, c-Fos or both were counted using microscopy to compare the proportion of activated histamine cells between control and laser activated mice.

Sleep scoring, analysis and statistics

We used a customized sleep detection script (Spike) for scoring sleep-wake states within the periods when laser stimulations were delivered. Sleep scoring protocol was followed as per previous publications from the Peever lab (Lyamin et al., 2016). The scorer was given 5 second epochs containing EEG and EMG signals. The scorer proceeded to define each epoch through the timeframe as wake, NREM sleep, and REM sleep based on cortical and muscle activity.

NREM sleep was characterized by high amplitude and low frequency brain activity and low to no muscle activity. REM sleep was characterized by high frequency, low amplitude cortical activity and absence of muscle activity with occasional twitches (Blumberg et al., 2020). The periods of wake were defined by high frequency and low amplitude cortical activity and presence of muscle tone.

Results:

Activation of the A11 dopamine projection to the TMN activates a higher proportion of histamine cells compared to control

Dopamine neurons of the A11 send inputs to the TMN region as shown through staining for dopamine and histamine neuronal markers tyrosine hydroxylase and histidine carboxylase respectively (Fig. 1a). First, we wanted to see whether A11 neurons excite wake-promoting histamine cells in the TMN. To do this, we injected AAV5-EF1α-DIO-ChETA-eYFP into the A11 region of Th-cre mice and then proceeded to optogenetically activate the (ChETA+) A11 terminals in the TMN (Fig. 1b) during a period when mice were mostly asleep for one hour. Mice were then perfused after a 90-minute time frame to allow for sufficient c-Fos expression in the brain. We stained neurons for markers of histamine (HDC) and neuronal activity (c-Fos) and proceeded to use the imaging program "Zen Zeis" to quantify the number of activated to non-activated neurons by marking cells that colocalize c-Fos (white) and HDC (red). Through this, we show that the activation of the A11 projections to the TMN activates significantly more histamine cells compared to control groups as shown by colocalization of c-Fos and histaminergic cells (Fig. 1c, d). This shows that the A11 to TMN dopaminergic projections activate histamine cells.

In vivo activation of A11 dopamine projections to the TMN consolidates wakefulness.

We next wanted to assess whether activation of A11 terminals in the TMN could generate wakefulness. To do so, we injected AAV5-EF1α-DIO-ChETA-eYFP into the A11 region of Th-cre mice and then proceeded to optogenetically activate the (ChETA+) A11 terminals in the TMN in vivo. The laser pulses were delivered at 20 Hz for 10 seconds every minute for a 3-hour period regardless of the sleep-wake state of the mice. This stimulation pattern allowed us to assess instantaneous and delayed changes caused by optogenetic activation. Optogenetic activation of the A11 terminals in the TMN produced an increased overall wakefulness amount in mice compared to their baseline which was not seen in control groups (Fig. 2a) (n=7, 2-way ANOVA, $p < 0.05$). We broke down the overall wakefulness amount into the total number of individual wake episodes and the duration of each episode. Interestingly, we found that activating the A11 terminals in the TMN increases the duration spent in each wake episode (Fig. 2b) and reduces the total number of wake episodes compared to baseline and controls (Fig. 2c) (n=7, 2-way ANOVA, $p < 0.05$) (Fig. 2c). Taken together, these behavioral data demonstrate that activation of the A11 dopamine input to the TMN does not initiate wake but rather prolongs wakefulness through activating wake-promoting histaminergic cells of the TMN.

Effects of A11 to TMN projections are robust enough to overcome sleep pressure

We next wanted to assess the strength of this neuronal circuit in the presence of physiological sleep pressure. Sleep pressure is the physiological push to fall asleep (NIOSH, 2020), which is modulated by sleep-promoting neuromodulators such as adenosine (Basheer et al., 2004). Th-cre mice expressing ChETA in dopamine neurons underwent a 3-day experiment (n=5). On day 1 mice were tethered only, and their sleep/wake cycle was noted. On day 2, mice were allowed to rest as they needed but the A11 projections in the TMN were laser stimulated for 3 hours following a pattern of 10-seconds on and 50 seconds off. On day 3, mice were sleep deprived following our protocol in their home cage for 2 hours during the day followed immediately by the same 3-hour laser protocol targeted at the TMN (Fig. 3a). The purpose of sleep deprivation was to induce physiological sleep pressure which increases with prolonged wakefulness which allows us to demonstrate if the pathway actively prevents sleep or maintains ongoing wakefulness under increased sleep pressure. Optogenetic activation of the A11 projections to the TMN during the rested trial shows significantly prolonged wakefulness compared to baseline which is consistent with our previous findings. However, in the presence of sleep pressure this pathway does not result in significant difference in wake amounts (Fig. 3b). Furthermore, we see a significant increase in the duration of individual wake episodes in both rested and sleep deprived trials (Fig. 3c), and although not yielding statistical significance, we show a decreasing trend in the total number of wake episodes in both rested and sleep deprived trials (Fig. 3d). These data suggest that the strength of the A11 to TMN pathway may be strong enough to overcome the physiological push to sleep, however further data is required.

Discussion:

Here we demonstrate the A11 dopamine neurons and the TMN form a functional circuit. We show that the wake promoting A11 dopamine neurons project onto and activate the histamine producing neurons of the TMN as shown through c-Fos staining. We further show through optogenetics that the activation of the A11 to TMN projections promotes wakefulness. This effect is seen through an increase in the duration spent in individual wake episodes, suggests that downstream targets of the A11 region promote wakefulness, which supports our initial hypothesis. Further experiments will be tailored to inhibiting this pathway using inhibitory opsin channels to yield a better understanding of the mechanism of this pathway. We want to assess whether this circuit can decrease wakefulness through increasing sleep pressure or by triggering sleep.

We further show that in the presence of sleep pressure this pathway increases the duration of each individual wake episode but does not significantly increase the overall wake amount. Interestingly, we see that the increase in the duration of each wake episode is very similar between rested and sleep deprived mice. This strongly suggests that the activation of this pathway can overcome the physiological pressure to sleep. However, more sleep deprivation subjects are required to gain statistical power and significance.

Additional research can focus on activating the A11 to TMN pathway and performing c-Fos staining on downstream regions of the TMN to assess the downstream effectors of the A11→TMN circuit. Activating this pathway could potentially through histamine release decrease activity of the pre-optic area (POA) which is a region that promotes sleep (Dong et al., 2023), and the connection between the TMN and POA could be a good candidate for future assessments. These further steps should also be coupled with fiber photometry and histamine sensors to measure histamine release onto this region in response to activation of the A11 dendrites in the TMN.

A better understating of the synaptic and circuit bases by which dopamine and histamine work together to maintain wakefulness has an important implication for patients suffering from neurological disorders associated with arousal dysfunction. An example is narcolepsy, where patients exhibit wake-instability and sleep attacks which take a toll on quality of life. By understating the circuitry involved in wake consolidation we can better understand the mechanism behind these disorders. Furthermore, current wake promoting drugs like modafinil, include the administration of systemic stimulants and amphetamines which have unwanted side effects. The work done towards this research opens the possibility to develop more selective pharmacological therapies targeting dopamine and histamine for promoting wakefulness which have major clinical implications.

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Figures:

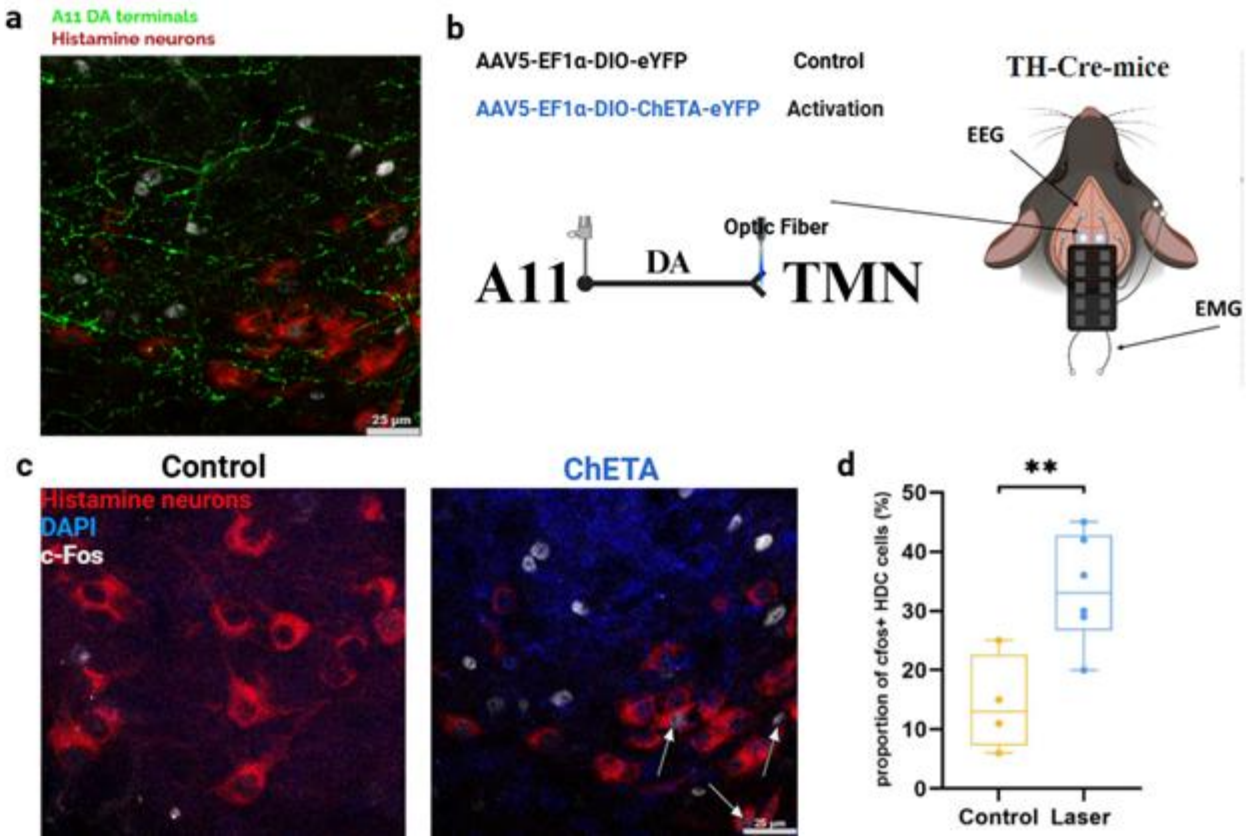

Fig. 1 Optogenetic activation of the A11 dopamine projections in the TMN activates histamine cells. (a) Dopaminergic A11 neurons send projections to the TMN (scale bar: 25µm). **(b)** AAV5-EF1α-DIO-ChETA-eYFP injections were made in the A11 region of the hypothalamus of Th-cre mice and optic fibers were placed above the TMN**. (c)** Activation of the A11 projections in the TMN (red showing HDC+) causes activation of TMN histamine neurons through increased expression of c-Fos (white) (scale bar: 25µm). **(d)** Proportion of c-Fos+ histamine neurons of the TMN after laser stimulation (n=6 ChETA) compared to controls (n=6 Control). (Unpaired t-test, $p < 0.05$).

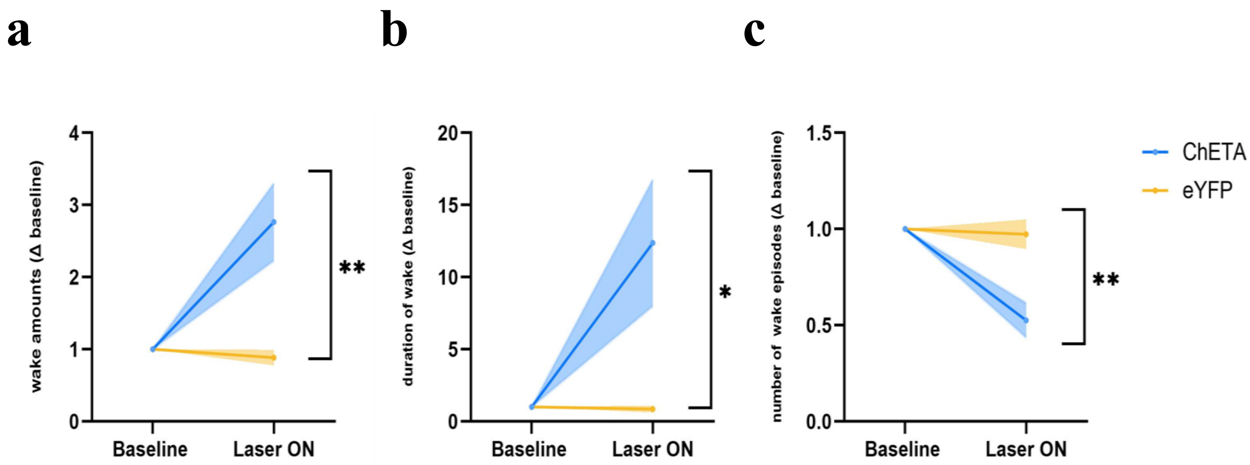

Fig. 2 Activation of the A11 dopamine projections in the TMN increases wakefulness. (a) Optogenetic activation of the A11 to TMN pathway causes mice to spend significantly more time in wakefulness compared to own baseline (blue) and control mice (yellow) (n=8 (ChETA), n=4 (Control), 2-way ANOVA, $p < 0.05$). These effects on the overall amount of wakefulness are due to Mice spending more time in each individual wake episode **(b)** and have reduced the overall number of wake episodes **(c)**

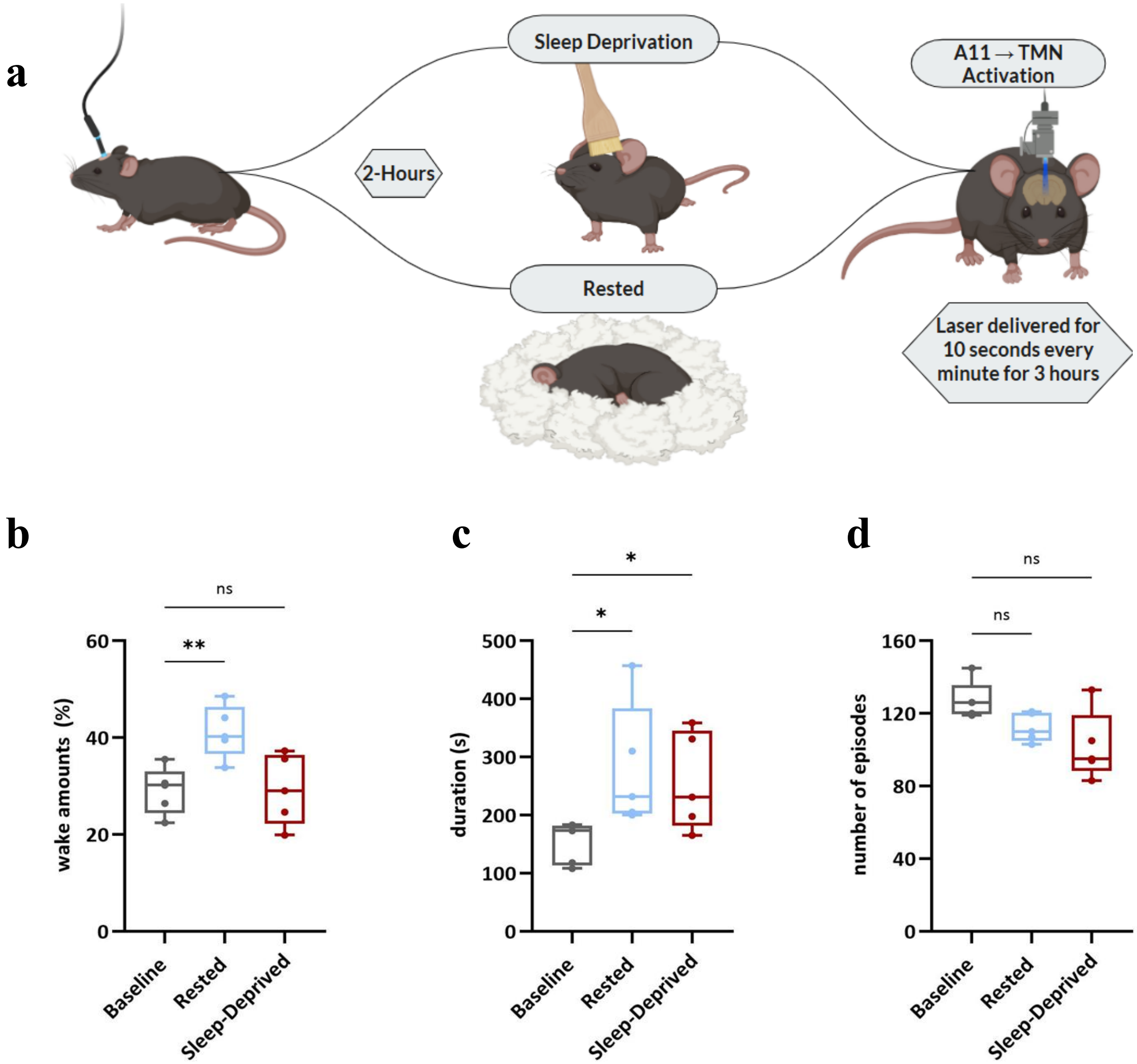

Fig. 3 Activation of A11 to TMN projections prolongs wakefulness under sleep pressure. **(a)** Protocol for rested, sleep deprivation and laser stimulation. **(b)** Rested mice (blue) show increased overall wakefulness when activating the A11 dendrites in the TMN compared to baselines (black) and sleep deprived mice (red) show no significant difference. **(c)** Both rested and sleep deprived mice show increased duration of individual wake episodes when given laser stimulation. (Repeated Measures ANOVA, p-value = 0.0956) **(d)** No statistically significant change in the number of wake episodes between laser stimulation days and baseline, although a downward trend is noticed (n=5).

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